Cambridge (CIE) A Level Maths: Pure 3

Your notes

Reciprocal Trigonometric Functions

Contents

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Your notes

Definitions of reciprocal trigonometric functions

What are the reciprocal trigonometric functions?

- There are three reciprocal trig functions corresponding to the three regular trig functions

1. SECANT (sec)

$$\sec x = \frac{1}{\cos x}$$

(UNDEFINED FOR VALUES OF x FOR WHICH $\cos x = 0$)

2. COSECANT (cosec)

$$\operatorname{cosec} x = \frac{1}{\sin x}$$

(UNDEFINED FOR VALUES OF x FOR WHICH $\sin x = 0$)

3. COTANGENT (cot)

$$\cot x = \frac{1}{\tan x}$$

(UNDEFINED FOR VALUES OF x FOR WHICH $\tan x = 0$; EQUAL TO ZERO FOR VALUES OF x FOR WHICH $\tan x$ IS UNDEFINED)

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- Cotangent can also be written in terms of sine and cosine

$$\cot x = \frac{\cos x}{\sin x}$$

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Examiner Tips and Tricks

- To solve equations with the reciprocal trig functions, convert them into the regular trig functions and solve in the usual way.
- Don't forget that both **tan** and **cot** can be written in terms of **sin** and **cos** (see the Worked Example below).

- You will sometimes see **csc** instead of **cosec** for cosecant.

Your notes

Worked Example

a) Prove that $\dfrac{1+\cot x}{1+\tan x} \equiv \cot x$.

b) Hence solve, in the interval $0 \le x \le 360°$,

the equation $\dfrac{1+\cot x}{1+\tan x} = 10$.

Give your answers to one decimal place.

a) $1+\cot x \equiv 1+\dfrac{\cos x}{\sin x} \equiv \dfrac{\sin x+\cos x}{\sin x}$

$1+\tan x \equiv 1+\dfrac{\sin x}{\cos x} \equiv \dfrac{\cos x+\sin x}{\cos x}$

REWRITE IN TERMS OF sin AND cos

SO $\dfrac{1+\cot x}{1+\tan x} \equiv \dfrac{\left(\dfrac{\sin x+\cos x}{\sin x}\right)}{\left(\dfrac{\cos x+\sin x}{\cos x}\right)} \equiv \dfrac{\cos x}{\sin x} \equiv \cot x$

$\dfrac{a/b}{a/c} = \dfrac{c}{b}$

b) WE NEED TO SOLVE $\cot x = 10$

$\dfrac{1}{\tan x} = 10$

$\tan x = \dfrac{1}{10}$

$x = 5.71°$

USE $\tan^{-1}$ ON YOUR CALCULATOR TO FIND THE PRINCIPAL VALUE OF x

OR $x = 180 + 5.71 = 185.71°$

SO $x = 5.7°$ OR $185.7°$ (1 d.p.)

THEN BE SURE TO FIND ANY OTHER VALUES IN THE REQUIRED INTERVAL

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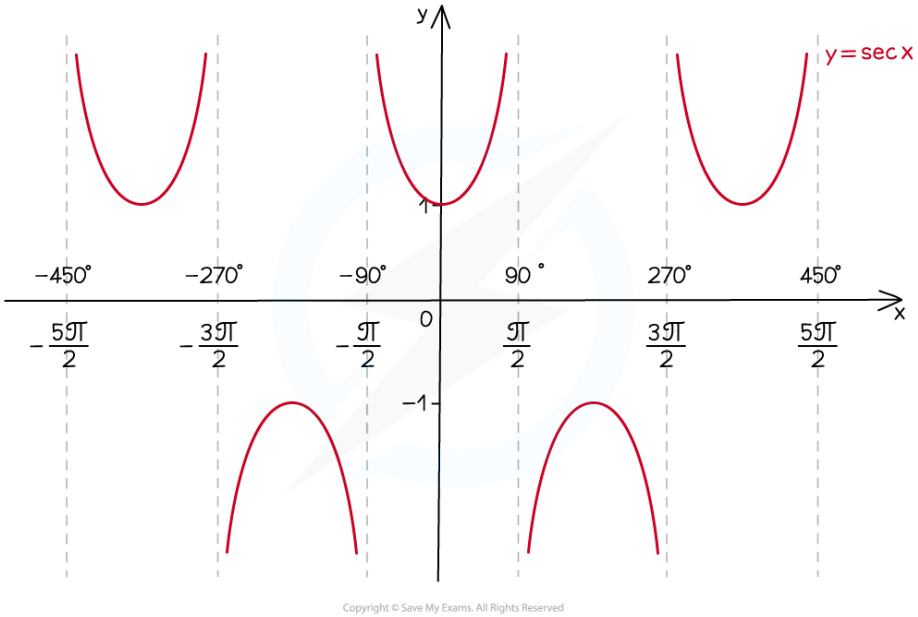
Graphs of reciprocal trigonometric functions

How do I sketch the graph of y = sec x?

- The graph of ***y* = sec*x*** looks like this:

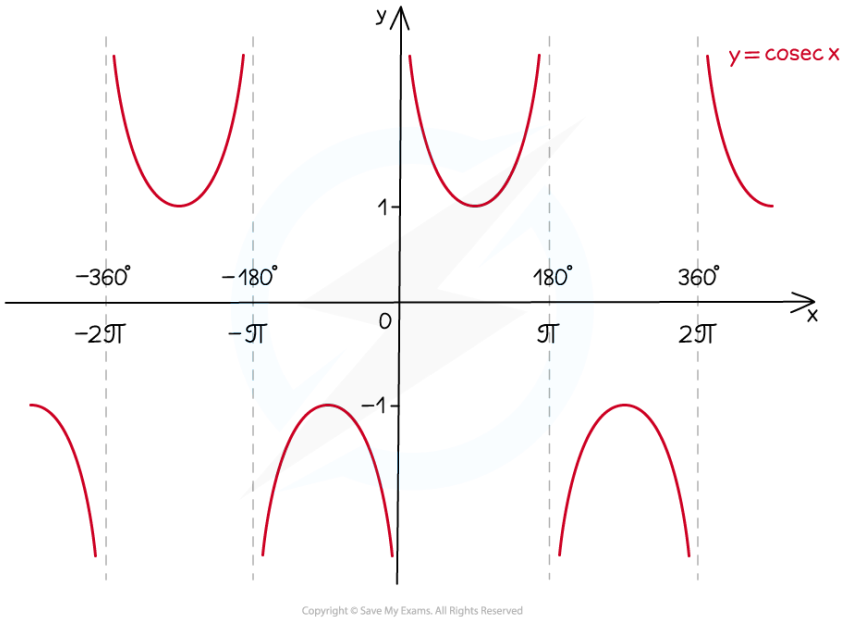


- ***y*-axis** is a **line of symmetry**
- has **period** (ie repeats every) **360°** or **2π** radians
- vertical **asymptotes** wherever **cos *x*= 0**
- **domain** is all *x* **except odd multiples of 90°** (90°, -90°, 270°, -270°, etc.)
- the **domain in radians** is all *x* **except odd multiples of π/2** (π/2, - π/2, 3π/2, -3π/2, etc.)
- **range** is ***y* ≤ −1 or *y* ≥ 1**

How do I sketch the graph of y = cosec x?

- The graph of ***y* = cosec *x*** looks like this:

Your notes



- has **period** (ie repeats every) **360°** or **2π** radians
- vertical **asymptotes** wherever **sin x= 0**
- **domain** is all *x* **except multiples of 180°** (0°, 180°, -180°, 360°, -360°, etc.)
- the **domain in radians** is all *x* **except multiples of π** (0, π, - π, 2π, -2π, etc.)
- **range** is ***y* ≤ -1 or *y* ≥ 1**

How do I sketch the graph of y = cot x?

- The graph of ***y* = cot *x*** looks like this:

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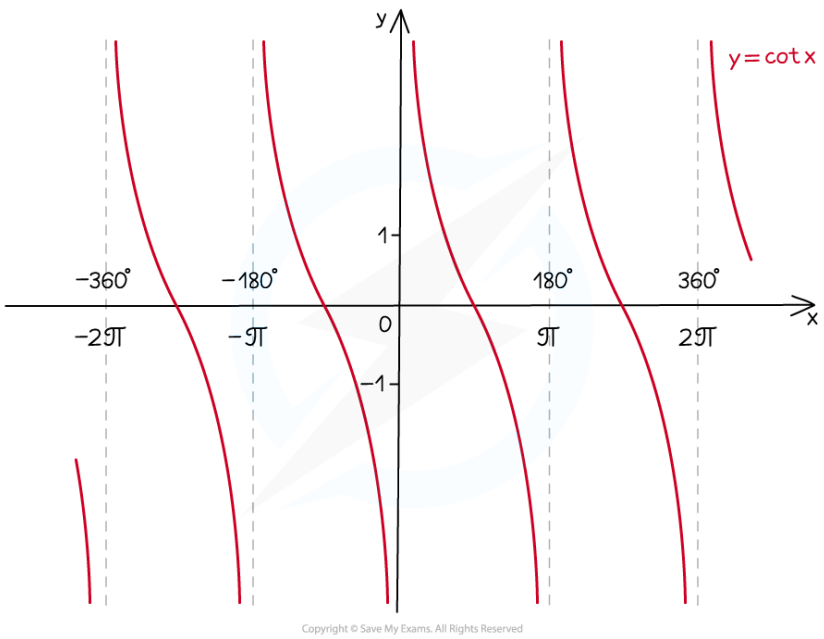


- has **period** (ie repeats every) **180°** or **π** radians
- vertical **asymptotes** wherever **tan *x* = 0**
- **domain** is all *x* **except multiples of 180°** (0°, 180°, -180°, 360°, -360°, etc.)
- the **domain in radians** is all *x* **except multiples of π** (0, π, - π, 2π, -2π, etc.)
- **range** is $y \in \mathbb{R}$ (ie **cot** can take *any* real number value)

Examiner Tips and Tricks

- Make sure you know the shapes of the graphs for **cos**, **sin** and **tan**.
- The shapes of the reciprocal trig function graphs follow from those graphs plus the definitions **sec = 1/cos**, **cosec = 1/sin** and **cot = 1/tan**

Worked Example

Your notes

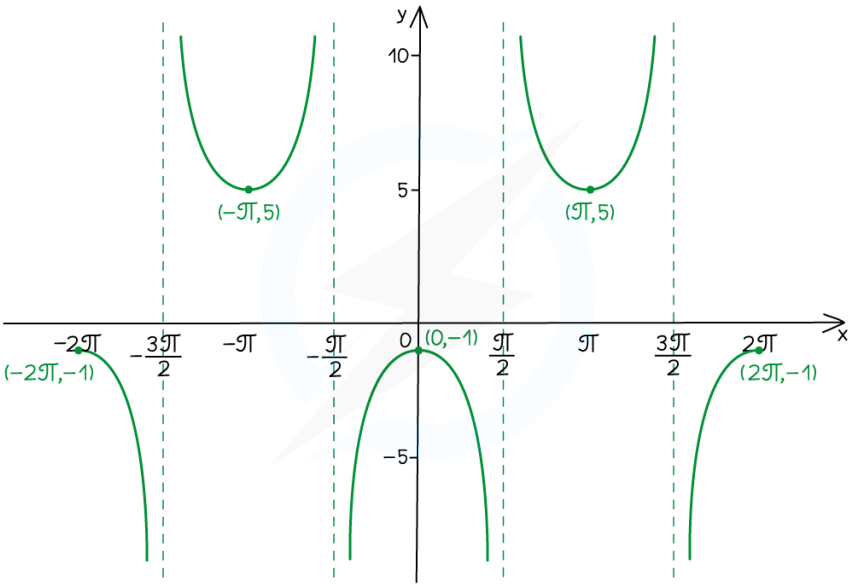
a) Sketch, in the interval $-2\pi \leq x \leq 2\pi$, the graph of $y = 2 - 3\sec x$.

b) Hence deduce the range of values of k for which the equation $2 - 3\sec x - k = 0$ has no solutions.

a)

THIS IS A TRANSFORMATION OF $y = \sec x$

- VERTICAL STRETCH, SCALE FACTOR 3
- THEN REFLECT IN x-AXIS
- FINALLY TRANSLATE VERTICALLY BY VECTOR $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$
- DON'T FORGET THE ASYMPTOTES



b) $2 - 3\sec x - k = 0$

$\Rightarrow 2 - 3\sec x = k$

FROM GRAPH, THE RANGE OF $y = 2 - 3\sec x$ IS $y \leqslant -1$ OR $y \geqslant 5$

SO THE EQUATION HAS NO SOLUTIONS WHEN $-1 < k < 5$

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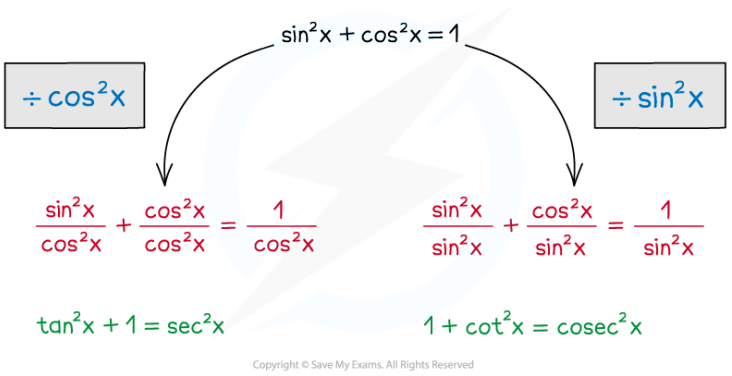
Identities with reciprocal trigonometric functions

What are the reciprocal trigonometric identities?

- There are two identities with **sec**, **cosec** and **cot** that you need to know and be able to use:
  - $\tan^2 x + 1 \equiv \sec^2 x$
  - $1 + \cot^2 x \equiv \operatorname{cosec}^2 x$
- These are not really 'new' identities – they can both be derived from $\sin^2 x + \cos^2 x \equiv 1$
  - To derive the identity for $\sec^2 x$ divide $\sin^2 x + \cos^2 x \equiv 1$ by $\cos^2 x$
  - To derive the identity for $\operatorname{cosec}^2 x$ divide $\sin^2 x + \cos^2 x \equiv 1$ by $\sin^2 x$



Examiner Tips and Tricks

- These identities are not given in the exam formulae booklet – make sure you know how to find them.
- They will be needed to prove some trigonometric identities and solve some trigonometric equations.

Worked Example

Your notes

Solve the equation $9\sec^2\theta - 11 = 3\tan\theta$,

in the interval $0 \le \theta \le 360°$.

Give your answers to 3 significant figures.

$(9\sec^2\theta - 9) - 2 = 3\tan\theta$

$\tan^2\theta + 1 = \sec^2\theta$
SO $\sec^2\theta - 1 = \tan^2\theta$

$9\tan^2\theta - 2 = 3\tan\theta$

THIS IS A QUADRATIC IN $\tan\theta$.
COMPARE $9y^2 - 3y - 2 = 0$

$9\tan^2\theta - 3\tan\theta - 2 = 0$

$(3\tan\theta - 2)(3\tan\theta + 1) = 0$

$\tan\theta = \frac{2}{3}$ OR $\tan\theta = -\frac{1}{3}$

USE $\tan^{-1}$ ON YOUR CALCULATOR TO FIND PRINCIPAL VALUES OF θ

IF $\tan\theta = \frac{2}{3}$

$\theta = 33.69$ OR $\theta = 180 + 33.69 = 213.69$

IF $\tan\theta = -\frac{1}{3}$

$\theta = -18.43$ OR $\theta = 360 - 18.43 = 341.57$

OR $= 180 - 18.43 = 161.57$

SO $= 33.7°, 162°, 214°, 342°$
(3 SIG. FIGS.)

BUT THEN BE SURE TO FIND ALL OTHER VALUES OF θ IN THE REQUIRED RANGE